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Send EMAIL

Aim

Develop an application to send Email.

Procedure:

Step 1 : File → New Project

Provide the application name (e.g., "Email") and click "Next".

Step 2 : Select the target Android devices

Select the minimum SDK to run the application. Click "Next".

Step 3 : Choose the activity for the application

By default, choose "Blank Activity". Click "Next".

Step 4 : Enter activity name and click "Finish".

Step 5 : Edit the program

Design the UI for entering recipient email, subject, and message.

Use an implicit intent (Intent.ACTION_SEND) in MainActivity.kt to trigger the email client with pre-filled details.

Step 6 : Run the application

Two ways to run the application:

1. Running through emulator
2. Running through mobile device

AndroidManifest.xml

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    package="com.example.email">

    <!-- Add internet permission -->
    <uses-permission android:name="android.permission.INTERNET" />

    <application
        android:allowBackup="true"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.SendEmail">
        <activity
            android:name=".MainActivity"
            android:exported="true">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />
                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>
    </application>

</manifest>
```

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    tools:context=".MainActivity">
    <!-- Header -->
    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:background="@color/purple_primary"
        android:padding="16dp"
        android:text="Send Email"
        android:textColor="@android:color/white"
        android:textSize="20sp"
        android:textStyle="bold"
        android:contentDescription="Send Email screen" />
    <!-- Email To Field -->
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginStart="16dp"
        android:layout_marginTop="16dp"
        android:text="To"
        android:textColor="@android:color/black"
        android:labelFor="@+id/to_edit_text" />
    <EditText
        android:id="@+id/to_edit_text"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginStart="16dp"
        android:layout_marginEnd="16dp"
        android:background="@drawable/edit_text_bottom_line"
```

```

        android:hint="Email address"
        android:importantForAutofill="yes"
        android:autofillHints="emailAddress"
        android:inputType="textEmailAddress"
        android:padding="16dp"
        android:minHeight="48dp"
        android:textSize="16sp" />
<!-- Subject Field -->
<TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginStart="16dp"
    android:layout_marginTop="16dp"
    android:text="Subject"
    android:textColor="@android:color/black"
    android:labelFor="@+id/subject_edit_text" />
<EditText
    android:id="@+id/subject_edit_text"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginStart="16dp"
    android:layout_marginEnd="16dp"
    android:background="@drawable/edit_text_bottom_line"
    android:hint="Subject"
    android:importantForAutofill="yes"
    android:autofillHints="subject"
    android:inputType="textCapSentences"
    android:padding="16dp"
    android:minHeight="48dp"
    android:textSize="16sp" />
<!-- Message Field -->
<TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginStart="16dp"
    android:layout_marginTop="16dp"
    android:text="Message"

```

```

        android:textColor="@android:color/black"
        android:labelFor="@+id/message_edit_text" />
<EditText
    android:id="@+id/message_edit_text"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginStart="16dp"
    android:layout_marginEnd="16dp"
    android:background="@drawable/edit_text_bottom_line"
    android:gravity="top"
    android:hint="Type your message"
    android:importantForAutofill="yes"
    android:autofillHints="emailBody"
    android:inputType="textMultiLine"
    android:minLines="3"
    android:minHeight="96dp"
    android:padding="16dp"
    android:textSize="16sp" />
<!-- Send Button -->
<Button
    android:id="@+id/send_button"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginStart="16dp"
    android:layout_marginTop="24dp"
    android:layout_marginEnd="16dp"
    android:background="@drawable/purple_button_background"
    android:text="Send"
    android:textAllCaps="false"
    android:textColor="@android:color/white"
    android:minHeight="48dp"
    android:textSize="16sp" />
</LinearLayout>

```

MainActivity.kt

```
package com.example.email

import android.content.Intent
import android.net.Uri
import android.os.Bundle
import android.widget.Button
import android.widget.EditText
import android.widget.Toast
import android.appcompat.app.AppCompatActivity

class MainActivity : AppCompatActivity() {

    private lateinit var toEditText: EditText
    private lateinit var subjectEditText: EditText
    private lateinit var messageEditText: EditText
    private lateinit var sendButton: Button

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)

        // Initialize UI components
        toEditText = findViewById(R.id.to_edit_text)
        subjectEditText = findViewById(R.id.subject_edit_text)
        messageEditText = findViewById(R.id.message_edit_text)
        sendButton = findViewById(R.id.send_button)

        // Set up send button click listener
        sendButton.setOnClickListener {
            sendEmail()
        }
    }
}
```

```

    }
}

private fun sendEmail() {
    val recipient = toEditText.text.toString().trim()
    val subject = subjectEditText.text.toString().trim()
    val message = messageEditText.text.toString().trim()

    // Basic validation
    if (recipient.isEmpty()) {
        Toast.makeText(this, "Please enter recipient email",
Toast.LENGTH_SHORT).show()
        return
    }

    try {
        // Try first with ACTION_SENDTO (preferred way for email)
        val emailIntent = Intent(Intent.ACTION_SENDTO).apply {
            data = Uri.parse("mailto:") // only email apps should handle
this
            putExtra(Intent.EXTRA_EMAIL, arrayOf(recipient))
            putExtra(Intent.EXTRA_SUBJECT, subject)
            putExtra(Intent.EXTRA_TEXT, message)
        }

        // Check if there's a matching activity
        if (emailIntent.resolveActivity(packageManager) != null) {
            startActivity(emailIntent)
            return
        }

        // Fallback to ACTION_SEND (more general, will show more apps)

```

```

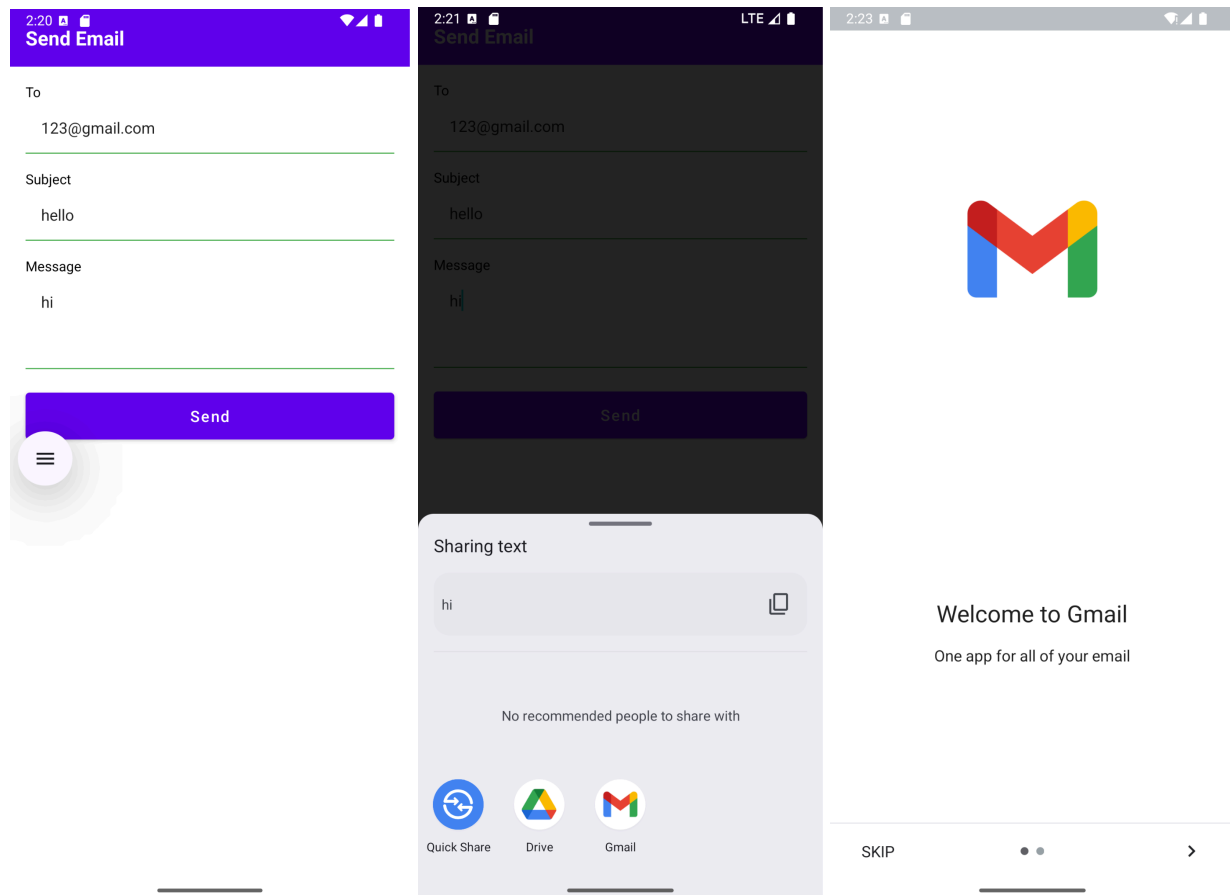
val sendIntent = Intent(Intent.ACTION_SEND).apply {
    type = "message/rfc822" // email mime type
    putExtra(Intent.EXTRA_EMAIL, arrayOf(recipient))
    putExtra(Intent.EXTRA_SUBJECT, subject)
    putExtra(Intent.EXTRA_TEXT, message)
}

// Use createChooser to always show options even if default is set
startActivity(Intent.createChooser(sendIntent, "Send email
using..."))

} catch (e: Exception) {
    // Show more detailed error message
    Toast.makeText(
        this,
        "No email app available. Error: ${e.message}",
        Toast.LENGTH_LONG
    ).show()
    e.printStackTrace()
}
}
}

```


Output



Result:

The Email application successfully opens the email client with pre-filled recipient, subject, and message, allowing the user to send an email via their preferred email app.